independent of the task they performed (see Figure 4). Moreover, participants seem to dehumanize robots to emotionless objects (i.e., lacking human nature) exclusively when a female robot performs analytical tasks or when a male robot performs social tasks (see Figure 5).

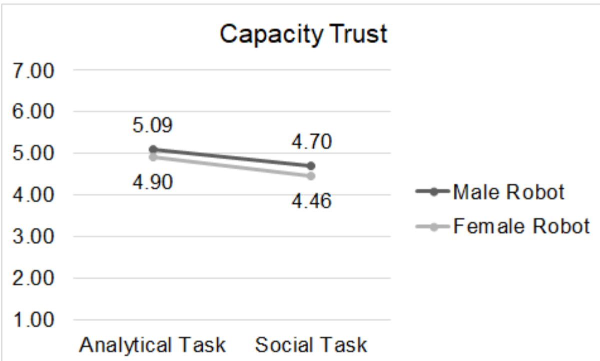

Figure 2: Robot gender vs. task type on capacity trust

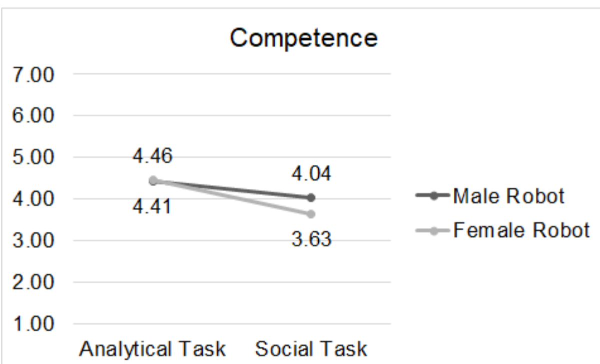

Figure 3: Robot gender vs. task type on competence

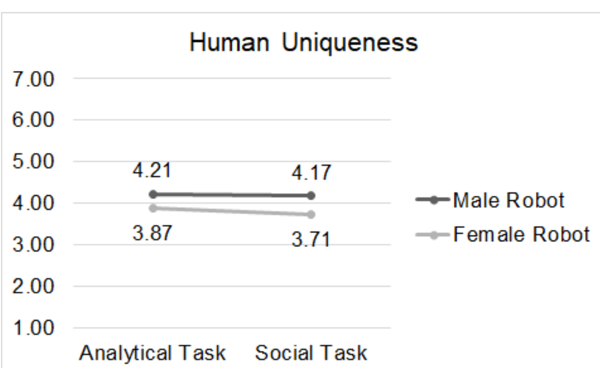

Figure 4: Robot gender vs. task type on human uniqueness

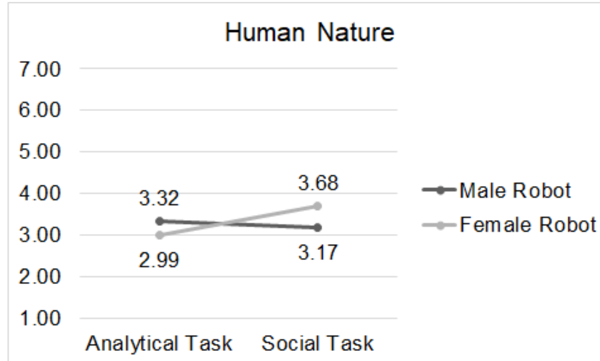

Figure 5: Robot gender vs. task type on human nature

4 GENERAL DISCUSSION

Our study expands current knowledge on appearance-task fit and gender inferences by investigating the effects of robot genderedness (male vs. female) and assigned task (analytical vs. social) on social perception, trust, and humanness.

Our data shows that people's evaluation of their trust in a robot only focuses on its capacity, but not on its morality, independent of its gender. These results show that trust evaluations of a robot cannot be linked to a robot's gender as we hypothesized (H2). Instead, our results indicate that trust in robots is more strongly associated with the performed task. Additionally, people perceive a robot to be more competent when it performs an analytical task compared to a robot performing a social task, independent of its gender. These results also contradict our hypothesis expecting an effect for robot gender on people's social perception of a robot (H1). It seems that, when associating gendered robots with specific tasks, the observed effects of gender stereotyping in both the psychology [3] and HRI [13] research seem to steer away from the genderedness of the embodiment towards the (perhaps also perceived gender-stereotypical) performed tasks –at least in terms of our social perception of and trust in such robots. An earlier study examining the relationship among occupational gender-roles, user trust and gendered robots also found no significant difference in the perception of trust in the robot's competency when considering the gender of the robot [5]. Similar findings have been reported in our HRI studies on gender-task fit [25], reporting that people are less willing to accept help from a robot with a typically female task (i.e., a social task).

These combined results on the dominant effects for task type, rather than for robot gender, are not completely surprising though. Prior research indeed shows that people in general hold more utilitarian perceptions on robots [9, 12, 15]. However, we also need to point to a potential limitation of our stimuli. Although the male robot was significantly perceived as more male than the female robot, its rating was leaning towards the female end of the gender scale. A similar observation was made for the social task, which was significantly perceived as less analytical than the analytical task yet its rating was leaning towards the analytical end of the scale. Future research should therefore not only explore different task descriptions but should also further investigate gendered appearances of robots or include of a gender-neutral robot as well as study subsequent (interaction) effects on people's social perception of and trust in such robots. Moreover, previous research in psychology [6] as well as HRI [34] shows interaction effects between the gender of the participant and that of the social other in terms of trust. Additional interaction effects between participant and robot gender have been found by [30] who reported a higher chance of an uncanny reaction to other-gender robots when that robot meets the gender expectations of warm females and competence males. Therefore, exploring interaction effects between the gender of the participant and the gender of robot in the context of gender-task fit might be promising as well.

Moreover, based on the literature in psychology, we expected that people's perceptions of a robot's humanness is a function of both robot gender and performed task (H3). This hypothesis was not supported with our data. However, we would like to discuss the observed trend in our data indicating a potential interaction effect between robot gender and performed task on a robot's perceived humanness. Following this trend, it seems that people dehumanize female robots (regardless of task performed) to animals lacking higher-level mental processes. Sexist responses to female robots have been reported in HRI research more generally [36, 40]. Additionally, people seem to dehumanize robots to emotionless objects only when gendered robots perform tasks contradicting the stereotypes of their gender. Research in social psychology has shown that women are dehumanized to both animals and objects [33], which is a precursor to aggressing against them [17]. Inserting the concept of gender into current debates regarding robot abuse (e.g., that mindless robots get bullied [23]) might offer alternative perspectives on these issues.

The field of social robotics aims to build robots that can engage in social interaction scenarios with humans in a natural, familiar, efficient, and above all intuitive manner [10]. The easiest way to deal with expectations of gendered robots and subsequent stereotypical impressions is to enhance people's social acceptance of gendered robots by tailoring their gender appearance to their intended task or application domain. Alternatively, robots might offer a unique potential to illuminate implicit bias in social cognition and challenging persisting gender-task stereotypes in society.

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